



清华大学  
Tsinghua University

Paper ID: PI38

# Boundary Inference of Load Scenarios in Multi-energy Parks Based on Statistical Learning

**Zhimeng WANG**

**Shenzhen International Graduate School, Tsinghua University**

**2023/7/29**



# Contents

- Background
- Problem Formulation
- Proposed Method
- Case Studies
- Conclusion



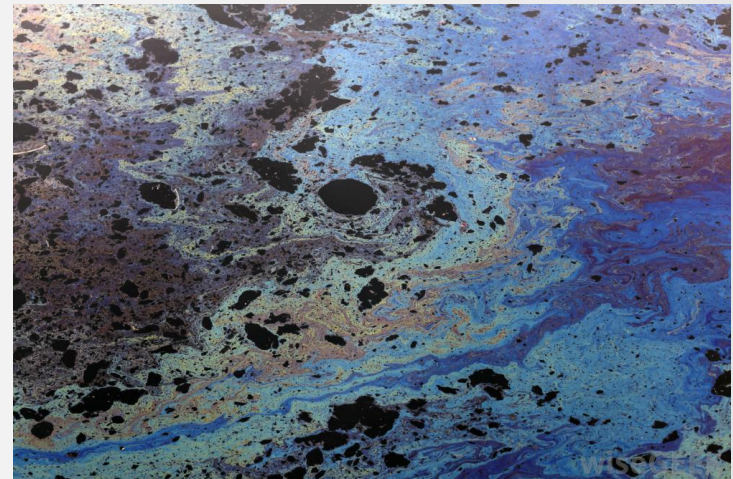
# Contents

- Background
- Problem Formulation
- Proposed Method
- Case Studies
- Conclusion

# Background



- Traditional fossil fuels have some unloved features.
  - Non-renewable: may be used up in a few decades.
  - Pollutable: have already led to many environmental issues such as acid rain, bad air quality and climate change.
  - Dangerous: when being used in improper ways.
- Traditional energy supply module is challenged, there must be a new and more proper way to supply energy.





# Background

- Traditional fossil fuels have some unloved features.
  - Non-renewable: may be used up in a few decades.
  - Pollutable: have already led to many environmental issues such as acid rain, bad air quality and climate change.
  - Dangerous: when being used in improper ways.
- Traditional energy supply module is challenged, there must be a new and more proper way to supply energy.
- Against such background, the concept of **multi-energy complementary park** came into being, which realizes the interconnection of various energy systems including electricity/gas/heat/cold, etc., and improves the comprehensive utilization efficiency of energy.
- However, various unexpected situations may occur during the operation of the multi-energy complementary park, resulting in various uncertainties, which make the optimal operation strategy of the multi-energy complementary parks face many difficulties. Modeling of uncertainties in multi-energy complementary parks is an urgent problem to be solved to handle the difficulties of operation of the systems.



# Contents

- Background
- Problem Formulation
- Proposed Method
- Case Studies
- Conclusion



# Problem Formulation

- Consider the following optimization model of a multi-energy complementary park:

$$\min_{x_i, Q_b, P_g} (C^{inv} + C^{ope})$$

**Objective function**

$$s.t. \quad C^{inv} = \sum_{i \in D} cc_i P_{i,g,\max} x_i$$

Investment cost definition

$$C^{ope} = \sum_{i=1}^D \sum_{t=1}^K c_i P_{i,g,t}$$

Operation cost definition

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{shed,t} = p_{0,t}$$

Power balance

$$0 \leq P_{i,g,t} \leq p_{0,t} \quad i = 1, \dots, D$$

Power generation limit

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t}$$

Power purchase limit

$$0 \leq P_{shed,t} \leq p_{0,t}$$

Load shedding limit

$$0 \leq |P_{i,g,t+1} - P_{i,g,t}| \leq \Delta P_{i,\max}$$

Ramping limit

$$0 \leq P_{i,g,t} \leq P_{i,g,\max} \quad i = 1, \dots, D$$

Power generation limit

# Problem Formulation

- Consider the following optimization model of a multi-energy complementary park;
- Taking uncertainty of load demand into consideration, a new model can be derived:

$$\min_{x_i, Q_b, P_g} (C^{inv} + C^{ope})$$

$$s.t. \quad C^{inv} = \sum_{i \in D} cc_i P_{i,g,\max} x_i$$

$$C^{ope} = \sum_{i=1}^D \sum_{t=1}^K c_i P_{i,g,t}$$

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{shed,t} = p_{0,t}$$

$$0 \leq P_{i,g,t} \leq p_{0,t} \quad i = 1, \dots, D$$

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t}$$

$$0 \leq P_{shed,t} \leq p_{0,t}$$

$$0 \leq |P_{i,g,t+1} - P_{i,g,t}| \leq \Delta P_{i,\max}$$

$$0 \leq P_{i,g,t} \leq P_{i,g,\max} \quad i = 1, \dots, D$$

**Objective function**

Investment cost definition

Operation cost definition

Power balance

Power generation limit

Power purchase limit

Load shedding limit

Ramping limit

Power generation limit

$$\min_{x_i, Q_b, P_g} \max_{u_p} (C^{inv} + C^{ope})$$

$$s.t. \quad C^{inv} = \sum_{i \in D} cc_i P_{i,g,\max} x_i$$

$$C^{ope} = \sum_{i=1}^D \sum_{t=1}^K c_i P_{i,g,t}$$

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{shed,t} = p_{0,t} \cdot u_{p,t}$$

$$0 \leq \sum_{i=1}^D P_{i,g,t} \leq p_{0,t} \cdot u_{p,t} \quad i = 1, \dots, D$$

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t} \cdot u_{p,t}$$

$$0 \leq P_{shed,t} \leq p_{0,t} \cdot u_{p,t}$$

$$0 \leq |P_{i,g,t+1} - P_{i,g,t}| \leq \Delta P_{i,\max}$$

$$0 \leq P_{i,g,t} \leq P_{i,g,\max} \quad i = 1, \dots, D$$



# Problem Formulation

- Consider the following four sets of constraints:

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{\text{shed},t} = p_{0,t}$$

Power balance

$$0 \leq P_{i,g,t} \leq p_{0,t} \quad i = 1, \dots, D$$

Power generation limit

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t}$$

Power purchase limit

$$0 \leq P_{\text{shed},t} \leq p_{0,t}$$

Load shedding limit

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{\text{shed},t} = p_{0,t} \cdot u_{p,t}$$

$$0 \leq \sum_{i=1}^D P_{i,g,t} \leq p_{0,t} \cdot u_{p,t} \quad i = 1, \dots, D$$

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t} \cdot u_{p,t}$$

$$0 \leq P_{\text{shed},t} \leq p_{0,t} \cdot u_{p,t}$$



# Problem Formulation

- Consider the following four sets of constraints:

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{\text{shed},t} = p_{0,t}$$

Power balance

$$0 \leq P_{i,g,t} \leq p_{0,t} \quad i = 1, \dots, D$$

Power generation limit

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t}$$

Power purchase limit

$$0 \leq P_{\text{shed},t} \leq p_{0,t}$$

Load shedding limit

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{\text{shed},t} = p_{0,t} \cdot \mathbf{u}_{p,t}$$

$$0 \leq \sum_{i=1}^D P_{i,g,t} \leq p_{0,t} \cdot \mathbf{u}_{p,t} \quad i = 1, \dots, D$$

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t} \cdot \mathbf{u}_{p,t}$$

$$0 \leq P_{\text{shed},t} \leq p_{0,t} \cdot \mathbf{u}_{p,t}$$

- Put the constraints into a compact form:

$$g(\mathbf{p}_0) \leq 0$$

Introduce uncertain variable  $\mathbf{u}_p$



$$g(\mathbf{p}_0, \mathbf{u}_p) \leq 0$$

# Problem Formulation

- Consider the following four sets of constraints:

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{\text{shed},t} = p_{0,t}$$

Power balance

$$0 \leq P_{i,g,t} \leq p_{0,t} \quad i = 1, \dots, D$$

Power generation limit

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t}$$

Power purchase limit

$$0 \leq P_{\text{shed},t} \leq p_{0,t}$$

Load shedding limit

$$\sum_{i=1}^D P_{i,g,t} + \frac{Q_{b,t}}{\Delta t} + P_{\text{shed},t} = p_{0,t} \cdot \mathbf{u}_{p,t}$$

$$0 \leq \sum_{i=1}^D P_{i,g,t} \leq p_{0,t} \cdot \mathbf{u}_{p,t} \quad i = 1, \dots, D$$

$$0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t} \cdot \mathbf{u}_{p,t}$$

$$0 \leq P_{\text{shed},t} \leq p_{0,t} \cdot \mathbf{u}_{p,t}$$

- Put the constraints into a compact form:

$$g(\mathbf{p}_0) \leq 0 \quad \xrightarrow{\text{Introduce uncertain variable } \mathbf{u}_p} \quad g(\mathbf{p}_0, \mathbf{u}_p) \leq 0$$

- In the problem of **boundary inference**,  $\mathcal{U}_{p,\epsilon}$ , the set of  $\mathbf{u}_p$ , should be found, and for  $\forall \mathbf{u}_p \in \mathcal{U}_{p,\epsilon}$ , there is:

$$\mathbb{P}(g(\mathbf{p}_0, \mathbf{u}_p) \leq 0) \geq 1 - \epsilon$$

Confidence level,  $0 < \epsilon < 1$ .



# Contents

- Background
- Problem Formulation
- Proposed Method
- Case Studies
- Conclusion

- The first step is to get the predicted load demand curve as input data:

$$\mathbf{p}_0 = \begin{bmatrix} p_{0,1} & p_{0,2} & \cdots & p_{0,K} \end{bmatrix}^T$$

- $K$  data points are input here as  $K$  continuous time points are considered at the same time.
- $K$  is usually set as 24 to count for the fact that there are 24 hours in each day.

# Proposed Method



Step 1 / Step 2 / Step 3

- Step 2 is to generate the alternative coefficients and choose among the coefficients based on calculation.

- Step 2 is to generate the alternative coefficients and choose among the coefficients based on calculation.
- The alternative coefficients are generated by choosing  $K$  numbers randomly from a normal distribution with a mean of 1. These  $K$  numbers are denoted by:

$$u^{(1)}, u^{(2)}, \dots, u^{(K)}$$

- Step 2 is to generate the alternative coefficients and choose among the coefficients based on calculation.
- The alternative coefficients are generated by choosing  $K$  numbers randomly from a normal distribution with a mean of 1. These  $K$  numbers are denoted by:

$$u^{(1)}, u^{(2)}, \dots, u^{(K)}$$

- Then sort these numbers from the smallest to the largest, the following list can be derived:

$$\hat{u}^{(1)}, \hat{u}^{(2)}, \dots, \hat{u}^{(K)} \quad \longleftarrow \text{Note the hats over the } u\text{'s.}$$

- Step 2 is to generate the alternative coefficients and choose among the coefficients based on calculation.
- The alternative coefficients are generated by choosing  $K$  numbers randomly from a normal distribution with a mean of 1. These  $K$  numbers are denoted by:

$$u^{(1)}, u^{(2)}, \dots, u^{(K)}$$

- Then sort these numbers from the smallest to the largest, the following list can be derived:

$$\hat{u}^{(1)}, \hat{u}^{(2)}, \dots, \hat{u}^{(K)} \quad \longleftarrow \text{Note the hats over the } u\text{'s.}$$

- Among the sorted numbers, the  $(K-s+1)$ th and the  $s$ th, i.e.,  $\hat{u}^{(K-s+1)}$  and  $\hat{u}^{(s)}$  would be chosen as the coefficients of the lower bound and the upper bound, respectively.

- Step 2 is to generate the alternative coefficients and choose among the coefficients based on calculation.
- The alternative coefficients are generated by choosing  $K$  numbers randomly from a normal distribution with a mean of 1. These  $K$  numbers are denoted by:

$$u^{(1)}, u^{(2)}, \dots, u^{(K)}$$

- Then sort these numbers from the smallest to the largest, the following list can be derived:

$$\hat{u}^{(1)}, \hat{u}^{(2)}, \dots, \hat{u}^{(K)} \quad \longleftarrow \text{Note the hats over the } u\text{'s.}$$

- Among the sorted numbers, the  $(K-s+1)$ th and the  $s$ th, i.e.,  $\hat{u}^{(K-s+1)}$  and  $\hat{u}^{(s)}$  would be chosen as the coefficients of the lower bound and the upper bound, respectively.

*What is  $s$  then? How to derive it?*

- $s$  can be calculated based on the following optimization problem [1][2]:

$$\begin{aligned} & \min_{s \in \mathbb{N}} s \\ \text{s.t. } & \sum_{j=k}^N \binom{N}{j} \left(\frac{1}{d}\right)^{N-j} \left(1 - \frac{1}{d}\right)^j \leq \frac{\alpha}{2d} \\ & 1 \leq s \leq K \\ & 0 \leq \sum_{i=1}^D P_{i,g,t} \leq p_{0,t} \cdot \hat{u}^{(K-s+1)} \\ & 0 \leq \sum_{i=1}^D P_{i,g,t} \leq p_{0,t} \cdot \hat{u}^{(s)} \\ & 0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t} \cdot \hat{u}^{(K-s+1)} \\ & 0 \leq \frac{Q_{b,t}}{\Delta t} \leq p_{0,t} \cdot \hat{u}^{(s)} \\ & 0 \leq P_{\text{shed},t} \leq p_{0,t} \cdot \hat{u}^{(K-s+1)} \\ & 0 \leq P_{\text{shed},t} \leq p_{0,t} \cdot \hat{u}^{(s)} \end{aligned}$$

[1] D. Bertsimas, V. Gupta, and N. Kallus, "Data-driven robust optimization," *Mathematical Programming*, vol. 167, pp. 235-292, 2018.

[2] P. Embrechts, A. Höing, and A. Juri, "Using copulae to bound the value-at-risk for functions of dependent risks," *Finance and Stochastics*, vol. 7, pp. 145-167, 2003.

# Proposed Method



Step 1 / Step 2 / Step 3

- The upper and lower boundaries of the uncertain fluctuation vector are obtained based on the coefficients corresponding to the upper and lower boundaries of the uncertain fluctuation vector in step 3.

- The upper and lower boundaries of the uncertain fluctuation vector are obtained based on the coefficients corresponding to the upper and lower boundaries of the uncertain fluctuation vector in step 3.
- According to  $s$  calculated in step 2,  $\hat{u}^{(K-s+1)}$  and  $\hat{u}^{(s)}$  are chosen as the lower bound and upper bound of the uncertain fluctuation vector, i.e., the following uncertainty set  $\mathcal{U}$  guarantees that more than  $(1-\alpha)$  of the samples can satisfy the security constraints  $g(\mathbf{p}, \mathbf{u})$  of the multi-energy complementary park at the level of  $\epsilon$ :

$$\mathcal{U}_{\epsilon} = \left\{ \mathbf{u} \in \mathbb{R}^K : \hat{u}^{(K-s+1)} \leq u_i \leq \hat{u}^{(s)} \quad i = 1, \dots, K \right\}$$



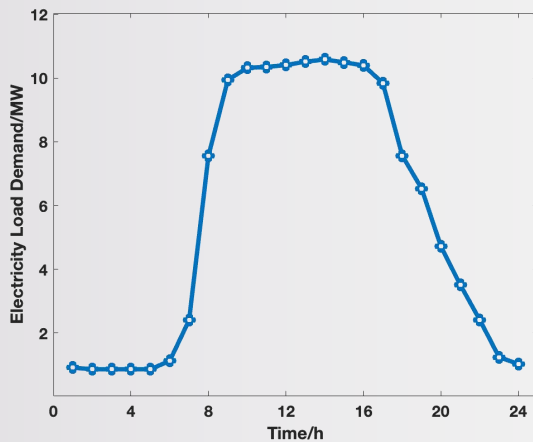
# Contents

- Background
- Problem Formulation
- Proposed Method
- Case Studies
- Conclusion



- In the numerical example, we assume all the data in the multi-energy park are recorded each hour, hence  $K=24$  points in a single day.
- The aforementioned method is first illustrated with a **toy example**, and results of the example is then **implemented** in a multi-energy complementary park to show the effect of the uncertainty set derived from the proposed method.

- In the numerical example, we assume all the data in the multi-energy park are recorded each hour, hence  $K=24$  points in a single day.



- In the numerical example, we assume all the data in the multi-energy park are recorded each hour, hence  $K=24$  points in a single day.

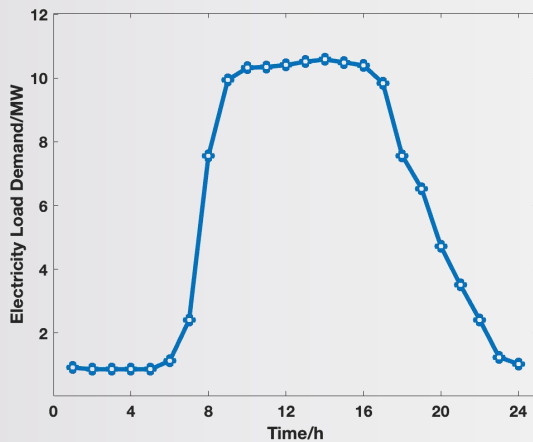


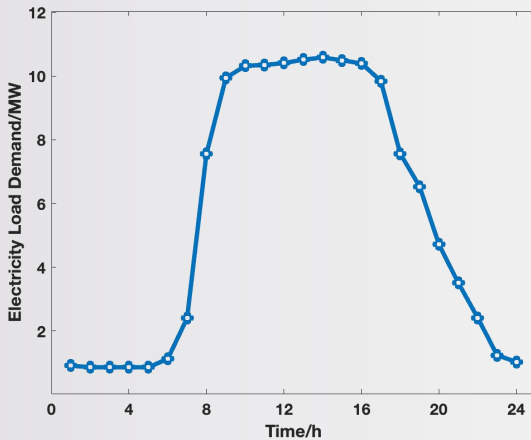
TABLE II  
THE GENERATED UNCERTAIN FLUCTUATION COEFFICIENTS

| Number       | $u^{(1)}$  | $u^{(2)}$  | $u^{(3)}$  | $u^{(4)}$  | $u^{(5)}$  | $u^{(6)}$  | $u^{(7)}$  | $u^{(8)}$  | $u^{(9)}$  | $u^{(10)}$ | $u^{(11)}$ | $u^{(12)}$ |
|--------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Coefficients | 1.317      | 1.162      | 1.128      | 0.883      | 0.352      | 0.962      | 0.801      | 1.882      | 0.944      | 0.159      | 0.845      | 0.585      |
| Number       | $u^{(13)}$ | $u^{(14)}$ | $u^{(15)}$ | $u^{(16)}$ | $u^{(17)}$ | $u^{(18)}$ | $u^{(19)}$ | $u^{(20)}$ | $u^{(21)}$ | $u^{(22)}$ | $u^{(23)}$ | $u^{(24)}$ |
| Coefficients | 0.341      | 1.984      | 1.101      | 0.637      | 0.261      | 1.153      | 0.683      | 1.319      | 1.487      | 0.412      | 1.302      | 1.219      |

TABLE III  
THE SORTED UNCERTAIN FLUCTUATION COEFFICIENTS

| Number       | $\hat{u}^{(1)}$  | $\hat{u}^{(2)}$  | $\hat{u}^{(3)}$  | $\hat{u}^{(4)}$  | $\hat{u}^{(5)}$  | $\hat{u}^{(6)}$  | $\hat{u}^{(7)}$  | $\hat{u}^{(8)}$  | $\hat{u}^{(9)}$  | $\hat{u}^{(10)}$ | $\hat{u}^{(11)}$ | $\hat{u}^{(12)}$ |
|--------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Coefficients | 0.159            | 0.261            | 0.341            | 0.352            | 0.412            | 0.585            | 0.637            | 0.683            | 0.801            | 0.845            | 0.883            | 0.944            |
| Number       | $\hat{u}^{(13)}$ | $\hat{u}^{(14)}$ | $\hat{u}^{(15)}$ | $\hat{u}^{(16)}$ | $\hat{u}^{(17)}$ | $\hat{u}^{(18)}$ | $\hat{u}^{(19)}$ | $\hat{u}^{(20)}$ | $\hat{u}^{(21)}$ | $\hat{u}^{(22)}$ | $\hat{u}^{(23)}$ | $\hat{u}^{(24)}$ |
| Coefficients | 0.962            | 1.101            | 1.128            | 1.153            | 1.162            | 1.219            | 1.302            | 1.317            | 1.319            | 1.487            | 1.882            | 1.984            |

- In the numerical example, we assume all the data in the multi-energy park are recorded each hour, hence  $K=24$  points in a single day.



$$s^* = 23$$

$$\hat{u}^{(K-s^*+1)} = \hat{u}^{(24-23+1)} = \hat{u}^{(2)} = 0.261$$

$$\hat{u}^{(s^*)} = \hat{u}^{(23)} = 1.882$$

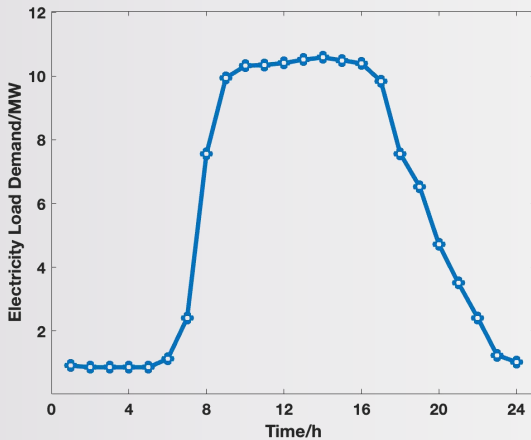
TABLE II  
THE GENERATED UNCERTAIN FLUCTUATION COEFFICIENTS

| Number       | $u^{(1)}$  | $u^{(2)}$  | $u^{(3)}$  | $u^{(4)}$  | $u^{(5)}$  | $u^{(6)}$  | $u^{(7)}$  | $u^{(8)}$  | $u^{(9)}$  | $u^{(10)}$ | $u^{(11)}$ | $u^{(12)}$ |
|--------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Coefficients | 1.317      | 1.162      | 1.128      | 0.883      | 0.352      | 0.962      | 0.801      | 1.882      | 0.944      | 0.159      | 0.845      | 0.585      |
| Number       | $u^{(13)}$ | $u^{(14)}$ | $u^{(15)}$ | $u^{(16)}$ | $u^{(17)}$ | $u^{(18)}$ | $u^{(19)}$ | $u^{(20)}$ | $u^{(21)}$ | $u^{(22)}$ | $u^{(23)}$ | $u^{(24)}$ |
| Coefficients | 0.341      | 1.984      | 1.101      | 0.637      | 0.261      | 1.153      | 0.683      | 1.319      | 1.487      | 0.412      | 1.302      | 1.219      |

TABLE III  
THE SORTED UNCERTAIN FLUCTUATION COEFFICIENTS

| Number       | $\hat{u}^{(1)}$  | $\hat{u}^{(2)}$  | $\hat{u}^{(3)}$  | $\hat{u}^{(4)}$  | $\hat{u}^{(5)}$  | $\hat{u}^{(6)}$  | $\hat{u}^{(7)}$  | $\hat{u}^{(8)}$  | $\hat{u}^{(9)}$  | $\hat{u}^{(10)}$ | $\hat{u}^{(11)}$ | $\hat{u}^{(12)}$ |
|--------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Coefficients | 0.159            | 0.261            | 0.341            | 0.352            | 0.412            | 0.585            | 0.637            | 0.683            | 0.801            | 0.845            | 0.883            | 0.944            |
| Number       | $\hat{u}^{(13)}$ | $\hat{u}^{(14)}$ | $\hat{u}^{(15)}$ | $\hat{u}^{(16)}$ | $\hat{u}^{(17)}$ | $\hat{u}^{(18)}$ | $\hat{u}^{(19)}$ | $\hat{u}^{(20)}$ | $\hat{u}^{(21)}$ | $\hat{u}^{(22)}$ | $\hat{u}^{(23)}$ | $\hat{u}^{(24)}$ |
| Coefficients | 0.962            | 1.101            | 1.128            | 1.153            | 1.162            | 1.219            | 1.302            | 1.317            | 1.319            | 1.487            | 1.882            | 1.984            |

- In the numerical example, we assume all the data in the multi-energy park are recorded each hour, hence  $K=24$  points in a single day.



$$s^* = 23$$

$$\hat{u}^{(K-s^*+1)} = \hat{u}^{(24-23+1)} = \hat{u}^{(2)} = 0.261$$

$$\hat{u}^{(s^*)} = \hat{u}^{(23)} = 1.882$$

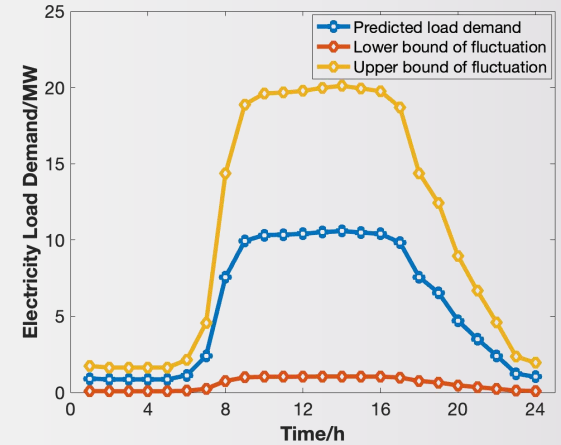


TABLE II  
THE GENERATED UNCERTAIN FLUCTUATION COEFFICIENTS

| Number       | $u^{(1)}$  | $u^{(2)}$  | $u^{(3)}$  | $u^{(4)}$  | $u^{(5)}$  | $u^{(6)}$  | $u^{(7)}$  | $u^{(8)}$  | $u^{(9)}$  | $u^{(10)}$ | $u^{(11)}$ | $u^{(12)}$ |
|--------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Coefficients | 1.317      | 1.162      | 1.128      | 0.883      | 0.352      | 0.962      | 0.801      | 1.882      | 0.944      | 0.159      | 0.845      | 0.585      |
| Number       | $u^{(13)}$ | $u^{(14)}$ | $u^{(15)}$ | $u^{(16)}$ | $u^{(17)}$ | $u^{(18)}$ | $u^{(19)}$ | $u^{(20)}$ | $u^{(21)}$ | $u^{(22)}$ | $u^{(23)}$ | $u^{(24)}$ |
| Coefficients | 0.341      | 1.984      | 1.101      | 0.637      | 0.261      | 1.153      | 0.683      | 1.319      | 1.487      | 0.412      | 1.302      | 1.219      |

TABLE III  
THE SORTED UNCERTAIN FLUCTUATION COEFFICIENTS

| Number       | $\hat{u}^{(1)}$  | $\hat{u}^{(2)}$  | $\hat{u}^{(3)}$  | $\hat{u}^{(4)}$  | $\hat{u}^{(5)}$  | $\hat{u}^{(6)}$  | $\hat{u}^{(7)}$  | $\hat{u}^{(8)}$  | $\hat{u}^{(9)}$  | $\hat{u}^{(10)}$ | $\hat{u}^{(11)}$ | $\hat{u}^{(12)}$ |
|--------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Coefficients | 0.159            | 0.261            | 0.341            | 0.352            | 0.412            | 0.585            | 0.637            | 0.683            | 0.801            | 0.845            | 0.883            | 0.944            |
| Number       | $\hat{u}^{(13)}$ | $\hat{u}^{(14)}$ | $\hat{u}^{(15)}$ | $\hat{u}^{(16)}$ | $\hat{u}^{(17)}$ | $\hat{u}^{(18)}$ | $\hat{u}^{(19)}$ | $\hat{u}^{(20)}$ | $\hat{u}^{(21)}$ | $\hat{u}^{(22)}$ | $\hat{u}^{(23)}$ | $\hat{u}^{(24)}$ |
| Coefficients | 0.962            | 1.101            | 1.128            | 1.153            | 1.162            | 1.219            | 1.302            | 1.317            | 1.319            | 1.487            | 1.882            | 1.984            |

- The derived uncertainty set is used in a specific multi-energy complementary park to verify its effectiveness.
- Candidate units include 4 dispatchable units, 2 nondispatchable units and 3 energy storage systems.

TABLE VIII  
PARAMETERS OF THE ENERGY STORAGE SYSTEMS

| Unit No. | Rated Power (MW) | Rated Energy (MWh) | Initial Investment Cost - Power (\$/MW) | Initial Investment Cost - Energy (\$/MWh) |
|----------|------------------|--------------------|-----------------------------------------|-------------------------------------------|
| 1        | 1                | 6                  | 60,000                                  | 30,000                                    |
| 2        | 2                | 6                  | 30,000                                  | 30,000                                    |

TABLE VI  
PARAMETERS OF THE DISPATCHABLE UNITS

| Unit No. | Capacity (MW) | Levelized Operation Cost (\$/MWh) | Initial Investment Cost (\$/MW) |
|----------|---------------|-----------------------------------|---------------------------------|
| 1        | 6             | 120                               | 50,000                          |
| 2        | 5             | 110                               | 60,000                          |
| 3        | 4             | 100                               | 70,000                          |
| 4        | 3             | 90                                | 80,000                          |

TABLE VII  
PARAMETERS OF THE NONDISPATCHABLE UNITS

| Unit No. | Capacity (MW) | Levelized Operation Cost (\$/MWh) | Initial Investment Cost (\$/MW) | Type of Energy |
|----------|---------------|-----------------------------------|---------------------------------|----------------|
| 1        | 4             | -                                 | 150,000                         | WT             |
| 2        | 2             | -                                 | 90,000                          | PV             |

- The derived uncertainty set is used in a specific multi-energy complementary park to verify its effectiveness.
- Candidate units include 4 dispatchable units, 2 nondispatchable units and 3 energy storage systems.

TABLE VI  
PARAMETERS OF THE DISPATCHABLE UNITS

| Unit No. | Capacity (MW) | Levelized Operation Cost (\$/MWh) | Initial Investment Cost (\$/MW) |
|----------|---------------|-----------------------------------|---------------------------------|
| 1        | 6             | 120                               | 50,000                          |
| 2        | 5             | 110                               | 60,000                          |
| 3        | 4             | 100                               | 70,000                          |
| 4        | 3             | 90                                | 80,000                          |

TABLE VIII  
PARAMETERS OF THE ENERGY STORAGE SYSTEMS

| Unit No. | Rated Power (MW) | Rated Energy (MWh) | Initial Investment Cost - Power (\$/MW) | Initial Investment Cost - Energy (\$/MWh) |
|----------|------------------|--------------------|-----------------------------------------|-------------------------------------------|
| 1        | 1                | 6                  | 60,000                                  | 30,000                                    |
| 2        | 2                | 6                  | 30,000                                  | 30,000                                    |

TABLE VII  
PARAMETERS OF THE NONDISPATCHABLE UNITS

| Unit No. | Capacity (MW) | Levelized Operation Cost (\$/MWh) | Initial Investment Cost (\$/MW) | Type of Energy |
|----------|---------------|-----------------------------------|---------------------------------|----------------|
| 1        | 4             | -                                 | 150,000                         | WT             |
| 2        | 2             | -                                 | 90,000                          | PV             |

TABLE IX  
PLANNING RESULTS AND TOTAL CONSTRUCTION COSTS GIVEN DIFFERENT UNCERTAINTY SETS

| Percentage of Lower Bounds | Percentage of Upper Bounds | Investment Decisions    |                            |                             | Costs (million \$) |
|----------------------------|----------------------------|-------------------------|----------------------------|-----------------------------|--------------------|
|                            |                            | Dispatchable Units (MW) | Nondispatchable Units (MW) | Energy Storage Systems (MW) |                    |
| 0.5                        | 1.5                        | 15                      | 2                          | 0                           | 1.0815             |
| <b>0.261</b>               | <b>1.882</b>               | <b>18</b>               | <b>4</b>                   | <b>0</b>                    | <b>1.7420</b>      |
| 0.1                        | 1.9                        | 18                      | 4                          | 0                           | 1.7423             |

# Case Studies



## A Toy Example / Implementation

- The derived uncertainty set is used in a specific multi-energy complementary park to verify its effectiveness.
- Candidate units include 4 dispatchable units, 2 nondispatchable units and 3 energy storage systems.

TABLE VI  
PARAMETERS OF THE DISPATCHABLE UNITS

| Unit No. | Capacity (MW) | Levelized Operation Cost (\$/MWh) | Initial Investment Cost (\$/MW) |
|----------|---------------|-----------------------------------|---------------------------------|
| 1        | 6             | 120                               | 50,000                          |
| 2        | 5             | 110                               | 60,000                          |
| 3        | 4             | 100                               | 70,000                          |
| 4        | 3             | 90                                | 80,000                          |

TABLE VIII  
PARAMETERS OF THE ENERGY STORAGE SYSTEMS

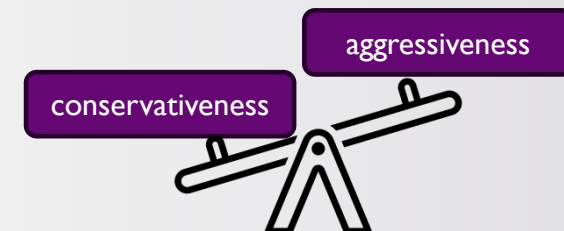
| Unit No. | Rated Power (MW) | Rated Energy (MWh) | Initial Investment Cost - Power (\$/MW) | Initial Investment Cost - Energy (\$/MWh) |
|----------|------------------|--------------------|-----------------------------------------|-------------------------------------------|
| 1        | 1                | 6                  | 60,000                                  | 30,000                                    |
| 2        | 2                | 6                  | 30,000                                  | 30,000                                    |

TABLE VII  
PARAMETERS OF THE NONDISPATCHABLE UNITS

| Unit No. | Capacity (MW) | Levelized Operation Cost (\$/MWh) | Initial Investment Cost (\$/MW) | Type of Energy |
|----------|---------------|-----------------------------------|---------------------------------|----------------|
| 1        | 4             | -                                 | 150,000                         | WT             |
| 2        | 2             | -                                 | 90,000                          | PV             |

TABLE IX  
PLANNING RESULTS AND TOTAL CONSTRUCTION COSTS GIVEN DIFFERENT UNCERTAINTY SETS

| Percentage of Lower Bounds | Percentage of Upper Bounds | Investment Decisions    |                            |                             | Costs (million \$) |
|----------------------------|----------------------------|-------------------------|----------------------------|-----------------------------|--------------------|
|                            |                            | Dispatchable Units (MW) | Nondispatchable Units (MW) | Energy Storage Systems (MW) |                    |
| 0.5                        | 1.5                        | 15                      | 2                          | 0                           | 1.0815             |
| <b>0.261</b>               | <b>1.882</b>               | <b>18</b>               | <b>4</b>                   | <b>0</b>                    | <b>1.7420</b>      |
| 0.1                        | 1.9                        | 18                      | 4                          | 0                           | 1.7423             |





# Contents

- Background
- Problem Formulation
- Proposed Method
- Case Studies
- Conclusion

# Conclusion



- A method of determining uncertainty sets in multi-energy complementary parks is proposed in this paper.
- By applying this method, the uncertainty sets can be derived flexibly since system operators can derive different uncertainty sets based on how much the safe operation of the system could be affected.
- Our method provides system operators with a reliable way to determine the uncertainty sets since no previous experience from system operators is needed, making the method easier to be popularized and utilized in various areas and scenarios.
- In the future, further comparison of this method and the other existing methods is necessary to further prove the significance and viability of the proposed method.



清华大学  
Tsinghua University

**Thanks for your attention.**

**Zhimeng WANG**

**Shenzhen International Graduate School, Tsinghua University**

**2023/7/29**